

CHIDVILAS KUMKAPALLA

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ACADEMIC PROFILE

IT graduate with hands-on experience in machine learning, data analytics, and generative AI frameworks. Proficient in Python, SQL, and MLOps pipelines, with hands-on experience developing data-driven systems ranging from RAG architectures to predictive analytics. Eager to advance expertise in cutting-edge Artificial Intelligence to solve complex, real-world engineering challenges.

EDUCATION

DR BR Ambedkar National Institute of Technology Jalandhar	2020-2024
Information Technology	CGPA: 7.32
Narayana Junior College, Gudivada.	2018-2020
Higher Secondary – JEE Mains 2020: 99.29 percentile	CGPA: 9.46

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Power BI
- **Machine Learning:** Scikit-learn, NLP, Regex, Computer Vision, Web Scraping
- **Agentic AI/Generative AI:** Prompt Engineering, RAG, LangChain, LangGraph, Fine-tuning
- **Core CS:** Data Structures & Algorithms, Database Systems, Cloud Computing.
- **Tools:** VS Code, Jupyter Notebook, Colab, Git, LaTeX, Salesforce, Streamlit, Hugging Face.

WORK & RESEARCH EXPERIENCE

LIVE DEMOS

Data Science with GenAI Intern (Nov 2025 – April 2026) 📍 Innomatics, Hyderabad

- **MEDHA!** – Architected and deployed an **Retrieval-Augmented Generation (RAG)** Q/A chatbot using Amazon Bedrock retrieve_and_generate API with Nova Pro LLM, OpenSearch vector store, and multi-turn conversational memory. Deployed on EC2 with UI, semantic document retrieval, & cited responses, reducing knowledge lookup time by eliminating manual doc search.
- **MindEase** – Built and deployed a production-ready domain-specific AI chatbot using Google Gemini API, implementing advanced prompt engineering, multi-turn conversational memory, and modular architecture; deployed on AWS EC2 with secure environment variable configuration, public IP accessibility, and background process management to ensure continuous system availability.
- **Flipkart Review Sentiment Analysis (MLOps Pipeline)** Developed a sentiment classification pipeline for Flipkart product reviews. Implemented experiment tracking for multiple models (Logistic Regression, LinearSVC) while logging **parameters, metrics, and artifacts** with MLflow. Optimized model performance using **F1-score (0.96)** and registered the best-performing model in the **MLflow Model Registry**. Orchestrated training with Prefect workflows.

Cloud Computing Intern

📍 1Stop, Bengaluru

- Deployed and configured a scalable VM on Microsoft Azure to host a web application, ensuring 99.9% uptime during the internship period. Gained practical experience in cloud infrastructure management, security group configuration, and full-stack application deployment on AWS.

PROJECTS

Beyond FOMO

[Live Demo](#)

Developed **Beyond FOMO**, an **Agentic AI** information assistant using **Python**, **LangChain**, **LangGraph**, and **Generative AI/LLMs**, orchestrating **10+ agent/tools** for intelligent information retrieval. Engineered **LangGraph pipelines** with **tool calling**, **SQLite conversational memory**, intent routing, and multiple **prompt-engineering strategies**, including summarization, FOMO-priority classification, opportunity detection, & **negative/toxic news filtering**.

Integrated **Hugging Face Whisper** for speech recognition, **Microsoft TrOCR** for *image-to-text extraction*, & Edge TTS for AI response. Built a **Streamlit-based frontend** supporting text, voice, & image inputs, with **web/news search**, **translation**, & real-time information retrieval.

Clinical Readmission Prediction

[Live Demo](#)

Built and deployed **Clinical Readmission Predictor**, an ML application using **Python**, **Scikit-learn**, and **Streamlit**, trained on 25K hospital encounter records. Performed comprehensive **EDA** and feature-target correlation analysis. Designed a robust **ML pipeline** with categorical encoding and **feature scaling**. Evaluated best **5 models** (Random Forest, XGBoost, DT, KNN & Logistic Regression) and applied **Optuna-based hyperparameter optimization (200 trials)** with **Stratified K-Fold Cross-Validation**. Deployed the final model as an **interactive Streamlit Webapp on Hugging Face** enabling real-time readmission risk prediction.

FarmSense360

Developed a platform to improve farming decisions with features such as a crop recommendation model for over 150 crops, crop price forecasting model using historical price data for over 30 different crops, an interactive community space, and diverse farming methods, leveraging various APIs to enrich the recommendation system. Crafted this system with React.js and Python in the backend, MongoDB database, and RESTful APIs, along with a UI built using Web-dev stack.

Data Analysis Projects

- **Energy Consumption Analysis** – Engineered a data analytics project using **MySQL** to generate business insights from global energy consumption datasets and optimized complex SQL queries to identify trends across **200+ countries**.
- **Power BI Weather and Climate Dashboard** – Developed a **dynamic Power BI dashboard** using real-time datasets from the **Indian Meteorological Department (IMD)** and designed interactive visualizations with filters for year, region, and district.
- **CO₂ Emissions Data Analysis** – Analyzed large datasets (**50,000+ rows**) using **Python (Pandas, NumPy)** and **SQL** to clean, transform, and structure data, reducing missing values by **98%** through automated preprocessing. Performed statistical analysis (Z-score, correlation, skewness) and built visualization dashboards.
- **AutoEV: Web-Scraped Electric Vehicle Analytics** – Developed a **Python web scraper using BeautifulSoup** to extract specifications for **1150+ EV models**, cleaned data using **Regex**, and performed EDA with visualization and outlier detection.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: **NASSCOM** – Advanced Data Science with Python, **University of Michigan** - Introduction to Data Science with Python

Languages: Telugu - Native, English (**IELTS Academic C1**), Hindi - Conversational.

Extra-Curricular: Member, Team Cultural Affairs, NIT Jalandhar – Led and organized multiple cultural programs, fostering teamwork and inclusivity.